KAIST
2026 Special Topics in Physics Quantum Electronic Devices I: Interferometers and Fractional Particles
Homework 3

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Contents

#1

(1a)

The boundary conditions for $V_1(x) = \alpha\delta(x)$ are

$$\psi(0^+) = \psi(0^-), \qquad \psi'(0^+) - \psi'(0^-) = \frac{2m\alpha}{\hbar^2}\psi(0). \tag{1}$$

With $\psi = e^{ikx} + re^{-ikx}$ for $x < 0$ and $\psi = te^{ikx}$ for $x > 0$, these become

$$1 + r = t, \qquad ik(t - 1 + r) = \frac{2m\alpha}{\hbar^2}\, t. \tag{2}$$

Defining $\gamma \equiv m\alpha/(\hbar^2 k)$ and substituting $r = t - 1$:

$$i(2t-2) = 2\gamma\, t \implies t(i-\gamma) = i \implies t = \frac{1}{1+i\gamma}, \quad r = \frac{-i\gamma}{1+i\gamma}. \tag{3}$$

Therefore,

$$T_1 = \frac{1}{1+\gamma^2}, \qquad R_1 = \frac{\gamma^2}{1+\gamma^2}. \tag{4}$$

(1b)

For $V_2(x) = \alpha[\delta(x) + \delta(x-d)]$, write

$$\psi = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0, \\ Ae^{ikx} + Be^{-ikx}, & 0 < x < d, \\ te^{ikx}, & x > d. \end{cases} \tag{5}$$

Boundary conditions at $x = 0$:

$$1 + r = A + B, \tag{6}$$
$$i(A - B - 1 + r) = 2\gamma(A + B). \tag{7}$$

Substituting $r = A + B - 1$ from (6) into (7):

$$i(A-1) = \gamma(A+B). \tag{8}$$

Boundary conditions at $x = d$:

$$Ae^{ikd} + Be^{-ikd} = te^{ikd}, \tag{9}$$
$$ik(te^{ikd} - Ae^{ikd} + Be^{-ikd}) = 2k\gamma\, te^{ikd}. \tag{10}$$

Using the continuity condition, the derivative condition simplifies to $iBe^{-ikd} = \gamma(Ae^{ikd} + Be^{-ikd})$, giving

$$B = \frac{\gamma A\, e^{2ikd}}{i - \gamma}. \tag{11}$$

Substituting (11) into (8) and rearranging:

$$(1+i\gamma)(1-A) = \gamma A \frac{i - \gamma + \gamma e^{2ikd}}{i-\gamma} \implies A = \frac{1+i\gamma}{(1+i\gamma)^2 + \gamma^2 e^{2ikd}}, \tag{12}$$

where we used $(1+i\gamma)^2 = 1+2i\gamma-\gamma^2$. The transmission amplitude is $t = A+Be^{-2ikd} = Ai/(i-\gamma) = A/(1+i\gamma)$:

$$t = \frac{1}{(1+i\gamma)^2 + \gamma^2 e^{2ikd}}. \tag{13}$$

Expanding $(1+i\gamma)^2 = (1-\gamma^2) + 2i\gamma$:

$$\begin{aligned} \left|(1+i\gamma)^2 + \gamma^2 e^{2ikd}\right|^2 &= \big[(1-\gamma^2) + \gamma^2\cos 2kd\big]^2 + \big[2\gamma + \gamma^2 \sin 2kd\big]^2 \\ &= (1+\gamma^2)^2 + \gamma^4 + 2\gamma^2\big[(1-\gamma^2)\cos 2kd + 2\gamma\sin 2kd\big]. \end{aligned} \tag{14}$$

Define θ by $\cos 2\theta = -\frac{1-\gamma^2}{1+\gamma^2}$ and $\sin 2\theta = \frac{2\gamma}{1+\gamma^2}$. Then

$$(1-\gamma^2)\cos 2kd + 2\gamma \sin 2kd = -(1+\gamma^2)\cos 2(kd+\theta), \tag{15}$$

and therefore

$$T_2 = \frac{1}{(1+\gamma^2)^2 + \gamma^4 - 2\gamma^2(1+\gamma^2)\cos 2(kd+\theta)}. \tag{16}$$

(1c)

Define the transfer matrix M by $\binom{A_L}{B_L} = M\binom{A_R}{B_R}$, where the wave on each side is $Ae^{iky} + Be^{-iky}$ with y measured from the scatterer. From (1a), $1 = (1+i\gamma)\,t$ and $r = -i\gamma\,t$, so the transfer matrix for a single delta barrier is

$$M = \begin{pmatrix} 1+i\gamma & i\gamma \\ -i\gamma & 1-i\gamma \end{pmatrix}. \tag{17}$$

The dynamical phase over distance d is encoded in the propagation matrix

$$U = \begin{pmatrix} e^{-ikd} & 0 \\ 0 & e^{ikd} \end{pmatrix}. \tag{18}$$

The total transfer matrix is $M_{\text{tot}} = MUM$. Computing the $(1,1)$ entry:

$$[M_{\text{tot}}]_{11} = (1+i\gamma)^2 e^{-ikd} + \gamma^2 e^{ikd} = e^{-ikd}\big[(1+i\gamma)^2 + \gamma^2 e^{2ikd}\big]. \tag{19}$$

Since $T_2 = 1/|[M_{\text{tot}}]_{11}|^2$ and $|e^{-ikd}| = 1$:

$$T_2 = \frac{1}{|(1+i\gamma)^2 + \gamma^2 e^{2ikd}|^2}, \tag{20}$$

which is identical to the result in (1b) before expanding absolute square.

(1d)

Refer to the Equation. 16, T_2 is maximized when $\cos 2(kd+\theta) = 1$, i.e.,

$$kd + \theta = n\pi, \quad n \in \mathbb{Z}. \tag{21}$$

At this maximum:

$$T_2^{\max} = \frac{1}{(1+\gamma^2)^2 + \gamma^4 - 2\gamma^2(1+\gamma^2)} = \frac{1}{[(1+\gamma^2) - \gamma^2]^2} = 1. \tag{22}$$

Since $T_1 = 1/(1+\gamma^2) < 1$ for any $\gamma \neq 0$, we have $T_2^{\max} > T_1$.

At the resonance condition, the two delta barriers form a cavity, and the particle undergoes multiple reflections between them. When the round-trip phase $2kd + 2\theta$ (dynamical phase $2kd$ and reflection phases 2θ) equals a multiple of 2π, the multiply-reflected waves interfere constructively in the forward direction and destructively in the backward direction, yielding perfect transmission—analogous to a Fabry–Pérot interferometer.